

# 1.2 kV Heavy-Duty Electric Vehicle Powertrains: System-Level Analysis and Holistic Efficiency Optimization of the Traction Inverter

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**ABSTRACT** The electrification of heavy-duty vehicles (HDVs) is essential for achieving zero-emission freight transport. Conventional 400–800 V powertrains face limitations in supporting megawatt-level fast charging and high continuous power due to excessive current, cable losses, and thermal stress. Recent advances in high voltage (i.e., 2 kV, 3.3 kV) silicon-carbide (SiC) power modules and charging standards such as the Megawatt Charging System (MCS, up to 1250 V) have enabled a new generation of 1200 V dc-link architectures for HDVs. This paper presents a comprehensive system-level evaluation of increasing the dc-link voltage from 800 V to 1200 V, covering the battery, inverter, and motor subsystems. At the inverter level, a unified virtual-prototyping with calorimetric loss validation framework is established to quantify conduction and switching losses, thermal behavior, and volume trade-offs. A holistic inverter optimization is then performed with respect to switching frequency, considering both inverter and motor harmonic losses to identify the efficiency-maximizing operating region. Experimental calorimetric measurements on a 250 kW, 1.2 kV dc-link inverter prototype based on SiC power modules confirm the analytical loss model within  $\approx 10\%$ , providing an experimental validation that is rarely reported for HDV traction inverters. Results show that 1200 V enables either a  $\sim 56\%$  loss reduction or a  $\sim 33\%$  cable-mass reduction in the charging system, maintaining compatibility with  $> 1$  MW fast charging, while the traction inverter prototype achieves efficiency of  $\sim 99.12\%$ , specific power density of 49.4 kW/L, and cubic power density of 20.83 kW/L. Accordingly, this paper provides experimentally supported design frameworks for next-generation high-voltage HDV powertrains.

**INDEX TERMS** High-voltage powertrain, traction inverter, heavy-duty electric vehicle, fast charging, dc-link voltage, insulation design, and harmonic losses.

## I. INTRODUCTION

Electrification of heavy-duty vehicles (HDVs) is one of the most effective strategies to reduce transport-related CO<sub>2</sub> emissions and to meet international climate goals such as those set by the Paris Agreement [1]. Therefore, in recent years, the automotive industry has accelerated the development of fully electric trucks to support the transition toward zero-emission heavy-duty transport [2].

Conventional HDV platforms typically operate with battery voltages up to 800 V [3]. However, the increasing power demand of long-haul applications and the need for shorter charging times have exposed the limitations of these voltage

levels [4]. The high currents required at 800 V cause excessive cable heating, increased conductor mass, and significant  $I^2R$  losses, which reduce overall efficiency and require complex cooling solutions for the charging infrastructure [5]. Moreover, at the inverter and motor levels, these high currents constrain both power density and thermal design margins [6]. To overcome these issues, higher-voltage architectures are being considered for next-generation HDVs, enabled by the availability of high-voltage SiC power modules (1.7–2.0 kV classes) and improved insulation technologies.

The development of new charging standards has further supported this transition. While the Combined Charging Sys-

tem (CCS) has become the de facto interface for commercial vehicles, the emerging Megawatt Charging System (MCS) supports voltages up to 1250 V and charging powers exceeding 1 MW [7]. Such capabilities are essential to satisfy regulatory requirements such as EU Regulation 561/2006, which mandates that professional drivers take a minimum 45-minute break after every 4.5 hours of driving [8]. This rule implies that a heavy-duty vehicle must be fully recharged within 45 minutes to maintain operational flexibility.

At the powertrain level, increasing the dc-link voltage reduces conduction losses and current stress across the drivetrain, thereby enabling higher efficiency and more compact designs [9], [10]. However, it also introduces challenges related to voltage-dependent stresses on dc-link capacitors and motor windings [11]. A comprehensive understanding of these trade-offs requires a system-level perspective that considers the battery, inverter, and traction motor, rather than analyzing each subsystem in isolation. Fig. 1 illustrates the coupled interactions, highlighting how dc-link voltage simultaneously affects charging current, inverter losses, and motor harmonic behavior. Despite recent progress, a unified and experimentally validated framework that quantifies these multi-domain interactions across voltage levels is still lacking, motivating the present research.

## A. RESEARCH GAP

Most existing studies focus on individual drivetrain subsystems such as the inverter [12], [13], [14], the motor [15], [16], [17], or the battery and charging system [18], [19], [20], [21], without assessing the complete drivetrain when the dc-link voltage is increased. Recent works have reviewed the transition from 400 V to 800 V architectures and analyzed their system-level benefits and implementation challenges for passenger electric vehicles [22], [5]. However, these analyses remain focused on light-duty applications, while the implications of further increasing the voltage beyond 1 kV in heavy-duty drivetrains have yet to be systematically investigated.

To fill this gap, this paper introduces a unified high-voltage (e.g. 1.2 kV) heavy-duty vehicle (HDV) assessment framework that integrates battery and charging behavior, inverter loss mechanisms, and motor harmonic losses. The proposed framework couples electro-thermal and electromagnetic interactions among drivetrain subsystems and is experimentally validated through calorimetric measurements and a loss separation method on a 250 kW, 1.2 kV dc-link traction inverter prototype, providing a consistent basis for quantifying design trade-offs in next-generation HDV powertrains.

## B. RESEARCH OBJECTIVE AND CONTRIBUTIONS

The objective of this paper is to quantify how increasing the dc-link voltage influences the charging performance, efficiency, power density, and harmonic losses of the complete electric drivetrain. The investigation encompasses the battery and charging system, traction inverter, and motor subsystem, as illustrated in Fig. 1.

The main contributions of this work are summarized as follows:

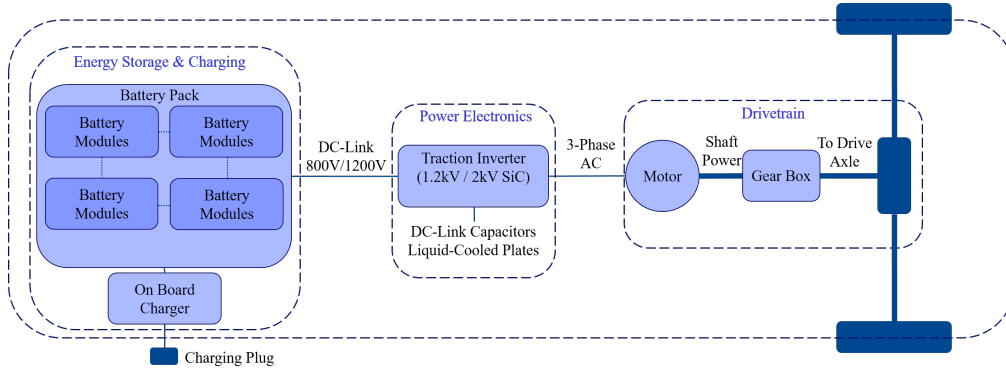
- 1) **Battery and Charging:** Assessment of 800 V and 1200 V architectures under fast-charging (i.e., MSC) conditions, quantifying charging current, cable losses, and achievable charging time.
- 2) **Traction Inverter:** Equal-power benchmarking of two 250 kW SiC systems, determining conduction and switching losses, dc-link capacitor requirements, cooling system, and resulting power-density trends.
- 3) **Motor Subsystem:** Analysis of winding and insulation adaptations along with copper, core, and harmonic losses using an impedance-based motor model.
- 4) **Loss Separation Method:** Development of a model to separate total inverter losses into conduction and switching components for experimental validation without requiring high-precision measurement probes.
- 5) **Holistic Optimization:** Implementation of a virtual-prototyping framework that identifies the switching frequency maximizing overall drivetrain efficiency by jointly considering inverter and motor harmonic losses.
- 6) **Experimental Verification:** calorimetric validation of the analytical loss model on a 250 kW SiC prototype inverter operating at 1200 V, confirming model accuracy.

Overall, this paper provides a unified and experimentally validated framework for assessing the transition from 800 V to 1200 V heavy-duty vehicle (HDV) powertrains, establishing design guidelines for high-voltage architectures that enhance efficiency, compactness, and fast-charging capability.

## II. SYSTEM REQUIREMENTS AND DESIGN BASIS

To provide a consistent basis for comparison, two dc-link voltage levels (800 V and 1200 V) are selected as representative of HDV powertrains. These values reflect the ongoing transition from existing 800 V platforms toward next-generation 1200 V architectures that support megawatt-class charging and improved drivetrain efficiency. Both systems are rated at 250 kW traction power. The inverters employ silicon-carbide (SiC) power modules with comparable chip technology, thermal characteristics, and cooling capability to ensure a fair comparison across voltage levels. Identical modulation indices and switching strategies are applied in both cases to maintain consistent operating conditions. The 800 V configuration utilizes 1.2 kV SiC devices, whereas the 1200 V setup adopts 2.0 kV modules, with both operated within equivalent thermal limits.

Table 1 summarizes the design parameters used as fixed inputs for the subsequent analyses, including loss modeling, inverter volume estimation, and motor harmonic-loss evaluation.



**FIGURE 1.** Schematic of the HDV traction system with battery, traction inverter, motor, and gearbox driving the axle.

**TABLE 1.** System Requirements and Design Parameters

| Parameter                        | 800 V      | 1200 V     |
|----------------------------------|------------|------------|
| Rated power, $P_{out}$ [kW]      | 250        | 250        |
| Dc-link voltage, $V_{dc}$ [V]    | 800        | 1200       |
| Phase Current, $I_{ph}$ [A]      | 294        | 196        |
| Input current, $I_{in}$ [A]      | 315        | 208        |
| Power factor, $p_f$              | 0.9-1      | 0.9-1      |
| Switching frequency, $f_s$ [kHz] | 10–25      | 10–25      |
| Semiconductor technology         | 1.2 kV SiC | 2.0 kV SiC |
| Cooling method                   | Liquid     | Liquid     |

### III. ANALYSIS OF BATTERY AND CHARGING SYSTEM

#### A. BATTERY PACK CONFIGURATION

The total energy stored in a traction battery is given by

$$E_{pack} = N_s N_p V_{cell} Q_{cell}, \quad (1)$$

where  $N_s$  and  $N_p$  represent the numbers of cells in series and parallel,  $V_{cell}$  is the nominal cell voltage, and  $Q_{cell}$  is the single-cell capacity. Increasing the dc-link voltage from 800 V to 1200 V raises the series cell count while reducing parallel branches, preserving approximately the same total energy capacity:

$$E_{pack,800V} \approx E_{pack,1200V}. \quad (2)$$

The resulting higher voltage lowers pack current, thereby reducing conduction losses in busbars and interconnects. At these elevated voltage levels, the design focus shifts toward insulation coordination between modules, which can be ensured through encapsulated housings, reinforced dielectric materials, and sufficient creepage and clearance distances [23]. These practices enable reliable operation under increased electric field stress across the battery pack.

#### B. FAST-CHARGING PERFORMANCE

The instantaneous charging current is defined as

$$I_{ch} = \frac{P_{ch}}{\eta_{ch} V_{dc}}, \quad (3)$$

where  $P_{ch}$  denotes the charging power,  $\eta_{ch}$  the charging efficiency, and  $V_{dc}$  the dc-link voltage. The corresponding cable conduction loss is

$$P_{loss} = I_{ch}^2 R_{cable}, \quad (4)$$

with  $R_{cable}$  representing the total resistance of the cable and connectors. For a constant charging power, increasing  $V_{dc}$  reduces the current and therefore the conduction losses according to

$$\frac{P_{loss,2}}{P_{loss,1}} = \left( \frac{V_{dc,1}}{V_{dc,2}} \right)^2, \quad (5)$$

indicating that an increase from 800 V to 1200 V lowers resistive losses to about 44% of their former value when the same conductor area is maintained.

Conversely, when the maximum cable current is fixed, the achievable charging power scales linearly with voltage:

$$P_{ch,max} = \eta_{ch} V_{dc} I_{max}. \quad (6)$$

Hence, a 1200 V system can deliver roughly 50% higher charging power than an 800 V counterpart under identical current limits. The associated charging time for an equal energy capacity decreases proportionally, shortening the duration by approximately one-third. This improvement enables complete recharging within the 45-minute rest period mandated by EU Regulation 561/2006 for commercial drivers [8].

#### C. THERMAL AND MECHANICAL CONSIDERATIONS

Building on the reduced charging current (for the same charging power), the associated losses are reduced, and the thermal stress on cables and connectors is alleviated. The required conductor cross-section can be approximated as

$$A = \frac{I_{ch}}{J_{max}}, \quad (7)$$

where  $J_{max}$  is the permissible current density. Since resistive heating scales with the square of the current ( $P_{loss} \propto I^2$ ), the decrease in current at higher dc-link voltage reduces thermal dissipation to about 45% of its previous level. This enables the use of cables with smaller cross-section and lower mass, typically decreasing total cable weight by 33%. The reduced

thermal load further simplifies cooling requirements and improves both operational safety and handling ergonomics during high-power charging.

#### D. QUANTITATIVE COMPARISON (1 MW TARGET)

Table 2 summarizes key parameters for 800 V and 1200 V systems when delivering 1 MW charging power, and additionally compares the 800 V case operated at the same current as the 1200 V system ( $I_{ch} = 877$  A). All cases assume  $\eta_{ch} = 0.95$  and a 600 kWh traction battery.

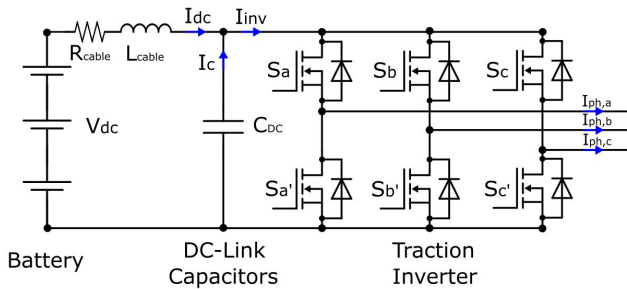
**TABLE 2. Comparison of 800 V and 1200 V Systems for 1 MW Charging ( $E_{pack} = 600$  kWh,  $\eta_{ch} = 0.95$ )**

| Parameter                | 800 V (1 MW) | 1200 V (1 MW) | 800 V (877 A) |
|--------------------------|--------------|---------------|---------------|
| $V_{dc}$ [V]             | 800          | 1200          | 800           |
| $I_{ch}$ [A]             | 1316         | 877           | 877           |
| $P_{ch}$ [kW]            | 1000         | 1000          | ~670          |
| $I^2R$ loss ratio        | 1.00         | 0.44          | 0.44          |
| $t_{ch}$ (600 kWh) [min] | 36           | 36            | ~54           |
| Cable mass trend         | higher       | ↓33%          | ↓33%          |
| MCS compliance           | partial      | full          | partial       |

Raising the dc-link voltage from 800 V to 1200 V lowers the charging current by about one-third and the associated  $I^2R$  losses by more than half for the same delivered power. When the current is held constant at 877 A, the 800 V system delivers roughly two-thirds of the power of its 1200 V counterpart, extending the full-charge duration from about 36 to 54 minutes. A 1200 V architecture, therefore, enhances charging efficiency, reduces cable mass, and aligns with the emerging MCS standard, supporting operation up to 1250 V.

#### IV. INVERTER BENCHMARK: VIRTUAL PROTOTYPING FRAMEWORK

This section compares two 2-level, 3-phase (2L–3Ph) SiC traction inverters, shown in Fig. 2, which differ only in their dc-link voltage levels (800 V and 1200 V).



**FIGURE 2. Two-level three-phase inverter schematic including dc-link capacitor and cable parasitics.**

Both configurations employ identical mechanical layouts, power-module packages, control hardware, and liquid-cooling arrangements. The comparison isolates the impact of dc-link voltage scaling on semiconductor losses, passive-component requirements, and overall power density under consistent design and thermal constraints.

#### A. ASSUMPTIONS AND COMPARISON METHODOLOGY

To ensure a fair and consistent comparison between the 800 V and 1200 V inverters, the following assumptions are applied:

- Identical module footprint (i.e., 62mm power module) and mechanical integration for both voltage levels.
- Equal total semiconductor losses at the rated operating point, verified through design-space exploration to maintain comparable cooling requirements.
- Maximum junction temperature limited to a specific value (i.e.,  $T_j \leq 125^\circ\text{C}$ ) to ensure thermal feasibility.
- Carrier-based sinusoidal PWM with balanced three-phase currents for dc-link current ripple assessment.
- Similar safety margins for device ratings to account for overshoot and reliability considerations.

These assumptions remove packaging, thermal, and control variations, allowing the comparison to isolate the effects of dc-link voltage scaling and associated current reduction.

#### B. CURRENT SCALING

For equal output power, the RMS phase current scales inversely with the dc-link voltage:

$$I_{1200} = I_{800} \frac{800}{1200}. \quad (8)$$

Accordingly, the 1200 V inverter operates at a proportionally lower current magnitude. However, the overall loss reduction is not directly proportional to the current decrease. Higher-voltage SiC devices employ thicker drift regions, which increase the specific on-resistance ( $R_{DS(on)}$ ) per unit area and partially offset the expected conduction-loss improvement. Moreover, switching losses tend to rise with device voltage class due to higher output capacitance and longer voltage-transition durations, especially at comparable gate-drive and  $dv/dt$  conditions. Therefore, both conduction and switching loss characteristics must be jointly evaluated to accurately quantify the net efficiency gain achievable through dc-link voltage scaling.

#### C. MODELING FRAMEWORK

A virtual-prototyping framework is established to evaluate inverter performance independently of any specific commercial module. Each design point is characterized by the switching frequency ( $f_{sw}$ ), active die area ( $A_{die}$ ), and device voltage class ( $U_b$ ). Semiconductor losses are calculated using technology-specific figures of merit (FOMs), while thermal feasibility is ensured by maintaining the junction temperature below its specified limit. The required dc-link capacitance and corresponding volume are derived from the dc-link ripple current characteristics. Finally, the total inverter volume and specific power density are obtained by assembling the volumes of the power modules, dc-link capacitors, and cooling system.



#### D. SEMICONDUCTOR LOSS MODELING

Semiconductor losses consist of conduction and switching components. The conduction loss per device is given by

$$P_c = \frac{1}{2\pi} \int_{\theta_{C1}}^{\theta_{C2}} r_{ds} i_{ds}^2(\theta) D(\theta) d\theta, \quad (9)$$

where  $D(\theta)$  is the duty ratio,  $i_{ds}(\theta)$  the instantaneous device current,  $\theta_{C1}$  to  $\theta_{C2}$  is conduction angle and  $r_{ds}$  is the on-state resistance, which can be estimated as

$$r_{ds} = \frac{k_R U_B^{\alpha_r}}{A_{die}}. \quad (10)$$

where  $A_{die}$  is active die area,  $k_R$ , and  $\alpha_r$  are technology-dependent coefficients, and  $U_B$  is the blocking voltage. This formulation captures the duty-cycle-weighted RMS conduction loss over one electrical period. In the conduction loss calculation, reverse conduction and body diode are neglected. These simplifications introduce opposing errors during reverse conduction and dead-time intervals that partially compensate each other [24], [25].

Switching losses can be split into two components: the zero-current term associated with the output capacitance and the overlap term related to  $dv/dt$  and  $di/dt$  during turn-on and turn-off transitions. The zero-current switching energy is approximated as

$$E_{sw,ZCS} = k_c U_b^{\alpha_c} A_{die} U_{dc}^2, \quad (11)$$

where  $k_c$ , and  $\alpha_c$  are technology-dependent coefficients,  $U_{dc}$  is the switching voltage that is the same as the dc-link voltage ( $V_{dc}$ ) in a 2-level inverter.

The overlap energy is expressed as

$$E_{sw,overlap} = \left( \frac{U_{dc} I_{sw}^2}{2 (di/dt)_{on}} + \frac{U_{dc}^2 I_{sw}}{2 (dv/dt)_{on}} \right) + \left( \frac{U_{dc} I_{sw}^2}{2 (di/dt)_{off}} + \frac{U_{dc}^2 I_{sw}}{2 (dv/dt)_{off}} \right), \quad (12)$$

where  $I_{sw}$  is the quasi-constant value of the switch current,  $dv/dt$  is the voltage slew rate, and  $di/dt$  is the current slew rate. Further, the overlap energy can be approximated by a low-order polynomial as

$$E_{sw,overlap}(I_{sw}, U_{dc}) \approx \alpha_1(U_{dc}) |I_{sw}| + \alpha_2(U_{dc}) I_{sw}^2. \quad (13)$$

where the coefficients  $\alpha_1$  and  $\alpha_2$  depend on the switching voltage, and voltage and current slew rates. The linear and quadratic current terms correspond to current-voltage overlap dominated by  $dv/dt$  and  $di/dt$  limitations, respectively. Using (12) and (13), the coefficients can be explicitly derived as

$$\alpha_1(V_{dc}) = \frac{V_{dc}^2}{2} \left[ \frac{1}{(dv/dt)_{on}} + \frac{1}{(dv/dt)_{off}} \right], \quad (14)$$

$$\alpha_2(V_{dc}) = \frac{V_{dc}}{2} \left[ \frac{1}{(di/dt)_{on}} + \frac{1}{(di/dt)_{off}} \right].$$

These relations show that slower voltage and current transitions (lower  $dv/dt$  and  $di/dt$ ) increase both coefficients,

resulting in higher switching losses, while faster transitions reduce  $\alpha_1$  and  $\alpha_2$  but worsen electromagnetic interference (EMI) and common-mode voltage stress. In this paper, the maximum values of  $dv/dt$  and  $di/dt$  are selected as 10 kV/ $\mu$ s and 10 kA/ $\mu$ s, respectively, and the coefficients  $\alpha_1$  and  $\alpha_2$  are estimated based on these limits. Alternatively, the coefficients  $\alpha_1$  and  $\alpha_2$  can be identified and validated experimentally through double-pulse measurements by fitting the measured  $E_{sw,overlap}(I_{sw})$  characteristics to (13) after subtracting the zero-current energy term given in (11).

Consequently, the total switching losses are obtained as

$$P_{sw} = \frac{f_{sw}}{2\pi} \int_{\theta_{SW1}}^{\theta_{SW2}} (E_{sw,ZCS} + E_{sw,overlap}) d\theta. \quad (15)$$

where  $\theta_{SW1}$  to  $\theta_{SW2}$  is switching angle.

The figure-of-merit and overlap-loss parameters for SiC MOSFETs are summarized in Table 3.

**TABLE 3. Figure-of-Merit Parameters[26] and Overlap-Loss Coefficients for SiC MOSFETs.**

| Parameter        | Value                | Unit                    |
|------------------|----------------------|-------------------------|
| $k_r$            | $7.2 \times 10^{-3}$ | $m\Omega mm^2 V^{-1.6}$ |
| $\alpha_r$       | 1.6                  | —                       |
| $k_c$            | $1.6 \times 10^4$    | $pF mm^{-2} V$          |
| $\alpha_c$       | —1                   | —                       |
| $\alpha_1(1200)$ | 144                  | $\mu J/A$               |
| $\alpha_2(1200)$ | 120                  | $nJ/A^2$                |
| $\alpha_1(800)$  | 64                   | $\mu J/A$               |
| $\alpha_2(800)$  | 80                   | $nJ/A^2$                |

#### E. IMPLEMENTATION OF THE LOSS MODELING ALGORITHM IN A TWO-LEVEL INVERTER

The conduction and switching intervals of the semiconductor devices are used to calculate the inverter losses according to (9) and (15). These intervals depend on the inverter topology and modulation scheme [25]. As summarized in Table 4, the conduction angles, duty ratios, and switching angles for each switch in a single inverter leg are defined for a positive half-cycle of the phase current.

**TABLE 4. Conduction and Switching Angles for Two-Level Inverters.**

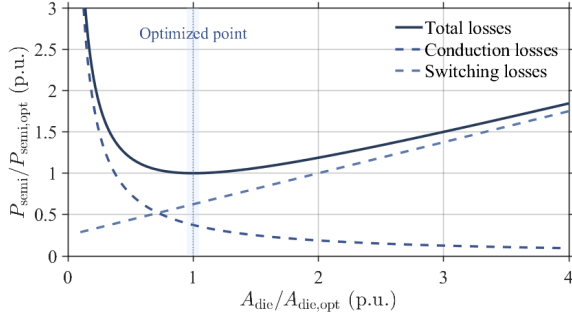
| Switch | Conduction Angle       | Duty ratio                           | Switching Angle        |
|--------|------------------------|--------------------------------------|------------------------|
| Upper  | $\phi$ to $\pi + \phi$ | $\frac{1+m_a \cdot \sin(\theta)}{2}$ | $\phi$ to $\pi + \phi$ |
| Lower* | $\phi$ to $\pi + \phi$ | $\frac{1-m_a \cdot \sin(\theta)}{2}$ | —                      |

\* means that the switch is in reverse conduction.

#### F. DEVICE RATING AND DIE-AREA OPTIMIZATION

Device voltage ratings are selected with safety factors of 1.5 and 1.66 to account for voltage overshoot, parasitic effects, and reliability margins [27], resulting in 1.2 kV devices for the 800 V system and 2.0 kV devices for the 1200 V system. Within a fixed package, the active die area is optimized to balance conduction and switching losses at a given switching frequency  $f_{sw}$  and voltage slew rate  $dv/dt$ . As illustrated in

Fig. 3, reducing the die area increases the on-state resistance  $r_{ds}$  and associated conduction losses, whereas enlarging the die area increases switching losses due to higher output capacitance. The intersection of these opposing trends defines



**FIGURE 3.** Conduction, switching, and total losses versus active die area at fixed  $f_{sw}$  and  $dv/dt$ . The optimal  $A_{die}$  minimizes total losses.

the optimal die area ( $A_{die,opt}$ ), corresponding to maximum efficiency while maintaining the junction temperature ( $T_j$ ) constraint.

#### G. THERMAL AND COOLING-SYSTEM ESTIMATION

A simplified thermal network is adopted, including the junction-to-case, case-to-cold-plate, and cold-plate-to-ambient thermal resistances. First, the junction-to-case thermal resistance decreases with increasing die area and is expressed as

$$R_{jc} = \frac{r_{th-die}}{A_{die}}, \quad (16)$$

where  $r_{th-die}$  is the die-specific thermal resistance coefficient. Second, the case-to-cold-plate thermal resistance is assumed to be determined by the device package and mounting conditions. Third, the required case-to-ambient thermal resistance to maintain the junction temperature constraint (i.e.,  $T_j \leq 125^\circ\text{C}$ ) is given by

$$R_{ca} = \frac{T_j - T_a}{P_{loss}} - (R_{jc-eq} + R_{cc-eq}), \quad (17)$$

where  $T_a$  is the ambient temperature,  $P_{loss}$  is the total semiconductor power loss, and  $R_{jc-eq}$  and  $R_{cc-eq}$  denote the equivalent junction-to-case and case-to-cold-plate thermal resistances, respectively.

Finally, the required cold-plate volume is estimated using the cooling-system performance index (CSPI) [28] as

$$V_h = \frac{1}{R_{ca} \cdot \text{CSPI}}. \quad (18)$$

#### H. DC-LINK CAPACITANCE ESTIMATION

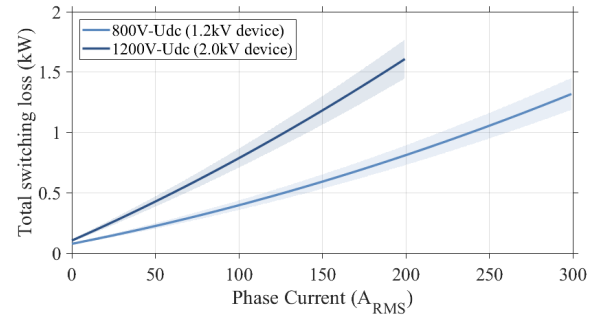
Assuming sinusoidal phase currents, the instantaneous dc-link capacitor current is expressed as

$$I_C(t) = \sum_{n=1}^{N_{ph}} S_n(t) I_n(t), \quad (19)$$

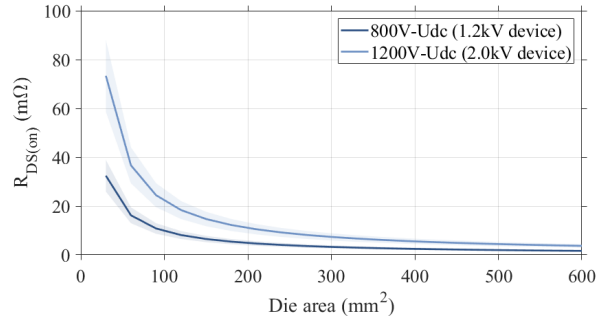
where  $S_n(t)$  denotes the switching functions represented by a double Fourier expansion [29]. The minimum required dc-link capacitance for a specified voltage ripple  $\Delta V^{pp}$  is calculated as

$$C_{bus} \geq \frac{\max\left(\int_0^{T_s} (I_C - I_{dc}) dt\right) - \min\left(\int_0^{T_s} (I_C - I_{dc}) dt\right)}{\Delta V^{pp}}. \quad (20)$$

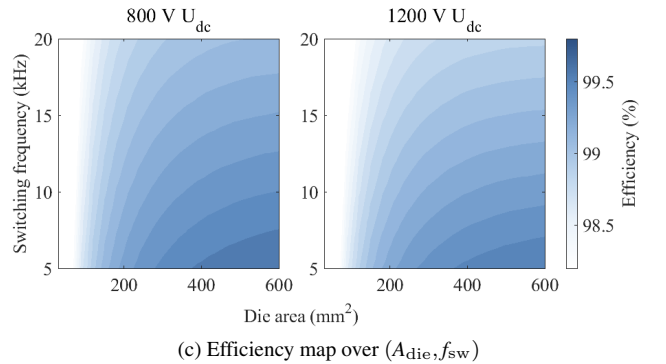
At constant power, increasing the dc-link voltage proportionally reduces the capacitor ripple current, thereby lowering the required capacitance; however, the capacitor volume per capacitance increases with dc-link voltage.



(a) Switching losses



(b) On-state  $r_{ds(on)}$



(c) Efficiency map over ( $A_{die}, f_{sw}$ )

**FIGURE 4.** Benchmark results at rated conditions: (a) switching losses at 10 kHz, (b) on-state resistance, (c) efficiency variation over die area and switching frequency.

#### I. DESIGN-POINT SELECTION AND EFFICIENCY MAPPING

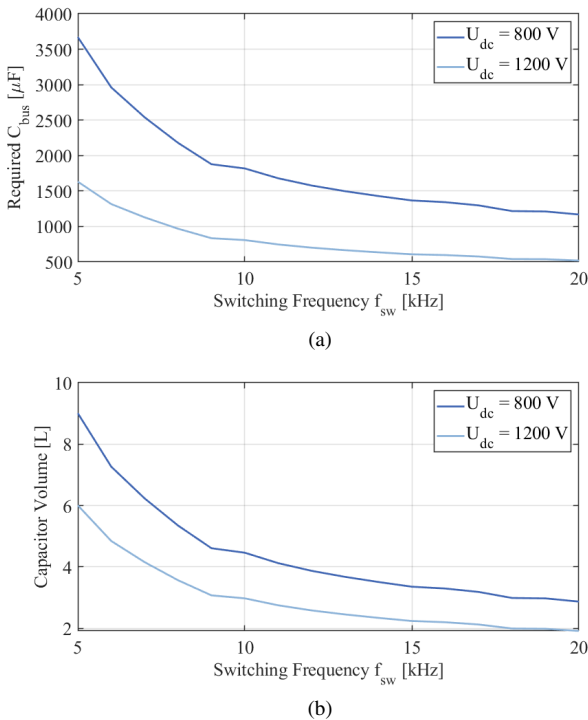
A parametric sweep across switching frequency and die area is performed for both voltage levels. Fig. 4 shows the switching-loss and on-state-resistance characteristics, along

with the corresponding efficiency distribution over the design space. The selected designs, achieving comparable total losses (at  $\eta \approx 99.2\%$ ), are summarized in Table 5.

**TABLE 5. Switching Frequencies and Die Areas of the Selected Operating Points.**

| Inverter | $f_{sw}$ (kHz) | $A_{die}$ (mm <sup>2</sup> ) | $\eta$ (%) |
|----------|----------------|------------------------------|------------|
| 800 V    | 14             | 300                          | 99.2       |
| 1200 V   | 10             | 300                          | 99.2       |

In addition to semiconductor-loss optimization, the dc-link capacitance requirements are evaluated as a function of switching frequency and voltage level. As shown in Fig. 5.a, increasing  $f_{sw}$  decreases the required capacitance since the charge ripple per switching period becomes smaller. Higher dc-link voltage levels further reduce ripple current, lowering capacitance demand for a given voltage ripple specification. To quantify the size impact, Fig. 5.b converts the capacitance into estimated capacitor volume based on representative volumetric energy densities. The 1200V design achieves roughly a 20% reduction in total capacitor volume compared with the 800 V system, illustrating a power-density benefit of higher dc-link operation.



**FIGURE 5. Effect of switching frequency on dc-link sizing for 800 V and 1200 V operation ( $m_a = 1$ ,  $\cos \varphi = 0.9$ ,  $P = 250$  kW). a) Required dc-link capacitance versus switching frequency. b) Estimated capacitor volume versus switching frequency.**

## J. THERMAL PARAMETERS AND VOLUME OF COOLING SYSTEM

Using  $r_{th,die} = 2.5$  °C/(W · mm<sup>2</sup>) and  $R_{ch} = 0.025$  °C/W per module, the equivalent thermal resistances and heatsink

requirements are calculated from (16)–(18). The cooling system can be further optimized as described in [30]; however, this is beyond the scope of this study. A liquid-cooled configuration with a coolant specific performance index (CSPI) of 40 W/(°C·L) is therefore assumed, yielding comparable cooling volumes for both voltage levels, as summarized in Table 6.

**TABLE 6. Thermal parameters and cold plate-to-ambient resistance.**

| Inverter | $T_j$ (°C) | $T_a$ (°C) | $R_{jc,eq}$ (°C/W) | $R_{cc,eq}$ (°C/W) | $R_{ca}$ (°C/W) | CSPI (W/°C·L) | Vol. (L)      |
|----------|------------|------------|--------------------|--------------------|-----------------|---------------|---------------|
| 800 V    | 125        | 25         | 0.0014             | 0.0083             | 0.0528          | 40            | $\approx 0.5$ |
| 1200 V   |            |            |                    |                    |                 |               |               |

## K. SUBSYSTEM VOLUMES AND SPECIFIC POWER DENSITY

The inverter volume is obtained by summing the contributions of the power modules, the dc-link capacitors, and the cooling system. The corresponding specific power density is expressed as

$$\rho_P = \frac{P_{rated}}{V_{pm} + V_{cap} + V_{cooling}}. \quad (21)$$

Table 7 summarizes the representative volume breakdown and resulting virtual-prototyping-based specific power densities. The difference in  $\rho_P$  primarily stems from the dc-link capacitor, whose volume decreases with increasing voltage level due to lower current ripple and capacitance demand. Consequently, the 1200 V inverter achieves a higher overall power density, benefiting from reduced current stress.

**TABLE 7. Component Volumes and Specific Power Density.**

| Inverter Voltage (V) | Power Module (L) | dc-link Capacitor (L) | Cold Plate (L) | Aux. System (L) | $\rho_P$ (kW/L) |
|----------------------|------------------|-----------------------|----------------|-----------------|-----------------|
| 800                  | 0.5              | 3.5                   | 0.5            | 0.5             | 50.0            |
| 1200                 | 0.5              | 3                     | 0.5            | 0.5             | 55.5            |

## L. SUMMARY OF FINDINGS

While the 800 V inverter exhibits lower  $r_{ds(on)}$  and switching energy per device, the higher current required for equal power output offsets these advantages. Under identical packaging, die area, and loss constraints, both voltage levels achieve comparable efficiency and thermal performance. Nevertheless, the 1200 V design benefits from reduced current stress and smaller dc-link capacitance, yielding a higher specific power density of 55.5 kW/L compared with 50.0 kW/L for the 800 V system.

## V. MOTOR LOSS EVALUATION

### A. WINDING AND INSULATION ADAPTATION

For a fair comparison between the 800 V and 1200 V systems, the same motor geometry and electromagnetic design are maintained. The higher dc-link voltage is accommodated by

reconnecting the existing parallel winding paths in series, while preserving the number of turns per slot and conductor dimensions. As a result, the torque constant, base speed, and magnetic loading remain unchanged, and the machine terminal voltage scales proportionally with the dc-link voltage.

### B. COPPER AND CORE LOSSES

The stator copper loss is

$$P_{cu} = 3 I_{rms}^2 R_s, \quad (22)$$

where  $R_s$  is the phase resistance at operating temperature. Since the same conductors are retained and the current per conductor is unchanged, copper loss remains effectively constant. However, higher insulation requirements at 1200 V typically mandate thicker groundwall and turn insulation [31], which reduces the slot fill factor and slightly increases  $R_s$ , leading to a marginal rise in  $P_{cu}$ .

Core losses remain essentially unaffected because the electrical frequency and air-gap flux density are identical in both voltage levels:

$$P_{core} = k_h f B^2 + k_e f^2 B^2, \quad (23)$$

where  $k_h$  and  $k_e$  are the hysteresis and excess-loss coefficients. This decomposition follows standard iron-loss formulations widely used in electric-machine design [32].

### C. HARMONIC-LOSS CHARACTERIZATION VIA IMPEDANCE-BASED LOSS FACTOR

Harmonic losses are evaluated using the machine's frequency-dependent impedance  $Z(j\omega)$ , obtained from small-signal LCR measurements or harmonic-injection tests over the relevant carrier and sideband frequencies [33], [34]. An example of the measured magnitude and phase response is shown in Fig. 6.

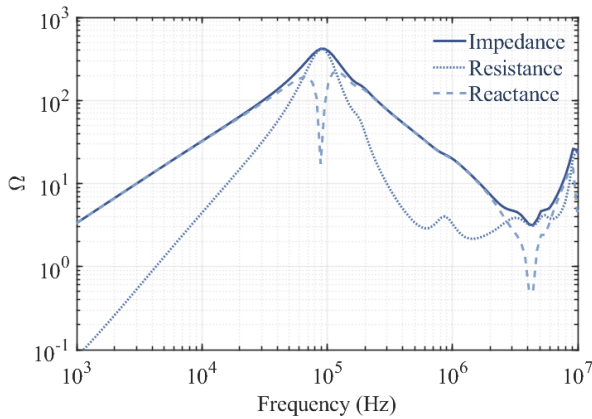


FIGURE 6. Measured frequency-dependent impedance  $Z(j\omega)$  of the 600 V<sub>dc</sub> rated machine.

From the measured impedance response, a voltage-based loss factor is extracted as

$$\Phi(\omega) = \frac{P_{add}(\omega)}{|V(\omega)|^2}, \quad (24)$$

where  $P_{add}(\omega)$  denotes the incremental loss associated with a small injected voltage at frequency  $\omega$ . The resulting frequency-dependent profile  $\Phi(\omega)$ , shown in Fig. 7, quantifies the additional loss contribution per unit squared harmonic voltage.

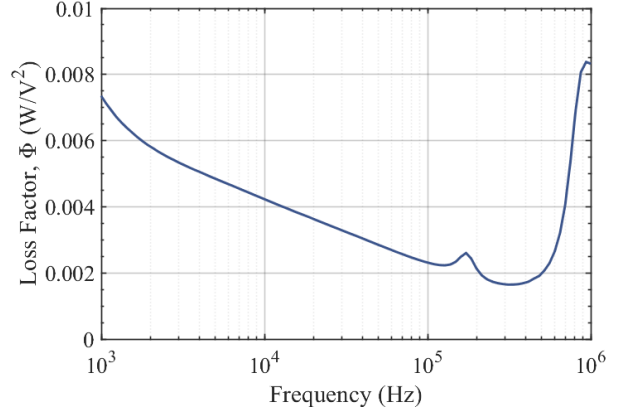


FIGURE 7. The derived loss factor  $\Phi(\omega)$  versus frequency.

### D. LOSS FACTOR SCALING WITH SERIES/PARALLEL RECONNECTION

Reconnecting parallel winding paths in series increases the effective phase inductance, altering the winding's high-frequency impedance behavior. Under this change, the loss factor  $\Phi(\omega)$  scales with the square of the inductance ratio as given:

$$\Phi(\omega)(V_{dc,2}) = \Phi(\omega)(V_{dc,1}) \frac{V_{dc,1}^2}{V_{dc,2}^2}, \quad (25)$$

### E. SYNTHESIS OF PWM HARMONIC LOSS

Under carrier-based sinusoidal PWM, the phase voltage can be expressed as a double-Fourier series composed of the fundamental component and carrier/sideband harmonics [29]:

$$\begin{aligned} \frac{v_{ph}(t)}{V_{dc}} &= \frac{A_{00}}{2} + \sum_{n=1}^{\infty} [A_{0n} \cos(n\omega_R t + \theta_R) + B_{0n} \sin(n\omega_R t + \theta_R)] \\ &+ \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} [A_{mn} \cos(\psi_{mn}) + B_{mn} \sin(\psi_{mn})], \end{aligned} \quad (26)$$

where  $\omega_R$  and  $\omega_C$  denote the fundamental and carrier angular frequencies, and  $\psi_{mn} = m(\omega_C t + \theta_C) + n(\omega_R t + \theta_R)$ . For fixed modulation index, the harmonic amplitudes scale linearly with the dc-link voltage.

Let  $\mathcal{H}$  denote the set of dominant carrier and sideband harmonics extracted from (26). Using the loss factor  $\Phi(\omega)$  of (25), the total additional harmonic loss is obtained by superposition:

$$P_{harm} = \sum_{h \in \mathcal{H}} \Phi(\omega_h) |V_h|^2, \quad (27)$$



where  $V_h$  is the phasor amplitude of the  $h$ -th voltage harmonic (e.g., obtained via the known Bessel-function coefficients for SPWM). This formulation enables a direct comparison of harmonic losses between 800 V and 1200 V systems by capturing both the voltage-scaling of harmonic amplitudes ( $|V_h| \propto V_{dc}$ ) and the voltage-scaled frequency dependence of the motor loss factor  $\Phi(\omega)$ . Accordingly, the harmonic loss at 1200 V can be expressed as

$$\begin{aligned} P_{\text{harm},1200} &= \sum_{h \in \mathcal{H}} \left( \frac{800^2}{1200^2} \Phi(\omega_{h,800}) \right) \left| \frac{1200}{800} V_{h,800} \right|^2 \\ &= P_{\text{harm},800}, \end{aligned} \quad (28)$$

indicating that identical harmonic losses are obtained for both dc-link voltage levels under the considered PWM conditions. This result explains why increasing the dc-link voltage does not inherently worsen PWM-induced motor losses when winding reconnection is properly applied.

#### F. OVERALL EFFECTS AND DESIGN IMPLICATIONS

The reconnection preserves the machine's fundamental electromagnetic characteristics, including the torque constant and base speed, but introduces a slight increase in phase resistance due to thicker insulation, resulting in a modest increase in copper losses. Core losses remain essentially unchanged. The reduced high-frequency loss factor  $\Phi(\omega)$  with higher dc-link voltage leads to similar harmonic losses under identical PWM schemes. Overall, the 1200 V configuration largely preserves motor efficiency, with only a slight increase in copper losses.

#### VI. EXPERIMENTAL VALIDATION FRAMEWORK: THE PROPOSED LOSS SEPARATION MODEL

Measuring individual loss components independently requires high-precision instrumentation and extensive post-processing across multiple operating points, while accounting for the applied PWM scheme. Therefore, this paper employs a calorimetric measurement setup, the schematic of which is shown in Fig. 8, to directly measure the total losses of the back-to-back inverters. Based on these measurements, a loss separation model is subsequently applied to estimate the conduction and switching loss components.

The total power loss was determined from the coolant temperature rise and the measured flow rate using

$$P_{\text{loss}} = \dot{m} c_p (T_{\text{out}} - T_{\text{in}}), \quad (29)$$

where  $\dot{m}$  is the coolant mass-flow rate,  $c_p$  its specific heat capacity, and  $T_{\text{out}}$  and  $T_{\text{in}}$  are outlet and inlet temperatures of the cold plate, respectively.

This measurement captures only the total power loss dissipated by the inverters mounted on the cold plate and does not distinguish between individual loss mechanisms. Consequently, analytical models for conduction and switching losses are formulated and parameterized, and their results are calibrated against the calorimetric measurements over different operating points to validate the analytical loss calculations.

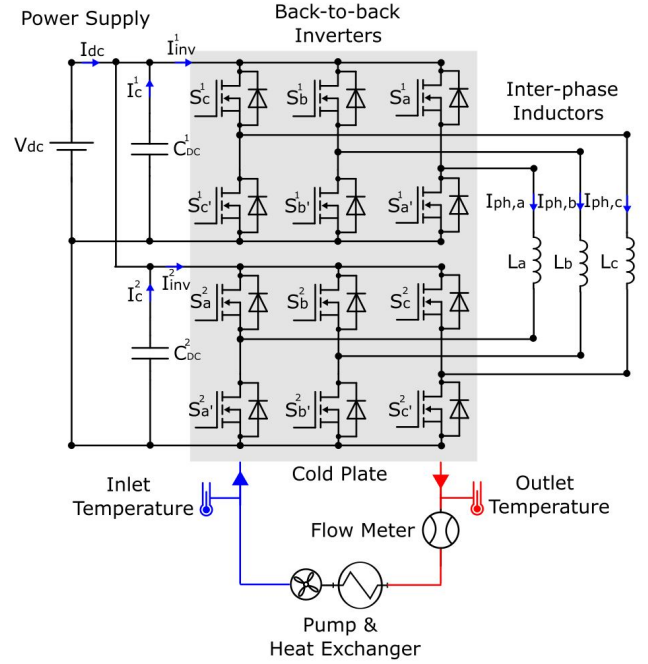


FIGURE 8. The back-to-back inverter with calorimetric setup.

#### A. CONDUCTION LOSS

Assuming a constant junction temperature in steady-state, the average conduction loss per leg is

$$P_{\text{cond}} = R_{\text{DS(on)}}(T_j) I_{\text{ph,RMS}}^2, \quad (30)$$

where  $R_{\text{DS(on)}}(T_j)$  is temperature dependent on-state resistance and  $I_{\text{ph,RMS}}$  is the RMS phase current of the three-phase inverter. The on-state resistance can be modeled as a linear function of the junction temperature:

$$R_{\text{DS(on)}}(T_j) = R_0 [1 + \alpha_R (T_j - T_{\text{ref}})], \quad (31)$$

where  $R_0$  is the resistance at the reference temperature  $T_{\text{ref}}$ , and  $\alpha_R$  is the temperature coefficient of resistance.

The junction temperature cannot be directly measured in typical calorimetric setups. It is therefore inferred from the total measured losses ( $P_{\text{tot}}$ ), the cold-plate temperature ( $T_{\text{cp}}$ ), and a factor  $\kappa_1$ , which relates the device junction to the cold-plate temperature:

$$T_j = T_{\text{cp}} + \kappa_1 P_{\text{tot}}. \quad (32)$$

The factor  $\kappa_1$  accounts for the effective thermal resistance network between the junction and the coolant, comprising spreading effects and sensor placement. This approach enables the estimation of thermal effects on the on-state resistance without requiring detailed thermal characterization or modeling as described in [35].

## B. SWITCHING LOSS

The total switching energy per device is expressed as the sum of the overlap energy and the capacitive energy of the output capacitance ( $C_{oss}$ ):

$$E_{sw}(I_{ph,RMS}) = E_{ol}(I_{ph,RMS}) + E_{Coss}(V_{dc}). \quad (33)$$

The overlap energy is represented by a low-order polynomial in the RMS current:

$$E_{ol}(I_{ph,RMS}) \approx \alpha_1 |I_{ph,RMS}| + \alpha_2 I_{ph,RMS}^2, \quad (34)$$

where  $\alpha_1$  and  $\alpha_2$  are empirical coefficients that account for the gate-drive network, device charge dynamics, and resulting  $dv/dt$  and  $di/dt$  characteristics. The capacitive energy contribution of  $C_{oss}$  during hard-switching transitions is modeled as

$$E_{Coss}(V_{dc}) \approx \frac{1}{2} C_{oss,eq} (V_{dc}) V_{dc}^2, \quad (35)$$

where  $C_{oss,eq}$  denotes the equivalent output capacitance, which can be extracted from zero-current switching tests:

$$E_{sw}(I_{ph,RMS} \approx 0) \approx \frac{1}{2} C_{oss,eq} V_{dc}^2. \quad (36)$$

The average switching power loss per leg is then given by

$$P_{sw} = f_{sw} (\alpha_1 I_{ph,RMS} + \alpha_2 I_{ph,RMS}^2 + C_{oss,eq} V_{dc}^2), \quad (37)$$

where  $f_{sw}$  denotes the switching frequency.

## C. PARAMETER CATEGORIZATION

Combining the above expressions, the total modeled loss per leg is

$$P_{tot,model} = P_{cond,leg}(T_j, I_{ph,RMS}) + P_{sw,leg}(I_{ph,RMS}, f_{sw}, V_{dc}), \quad (38)$$

where the conduction losses are temperature- and RMS current-dependent, and the switching losses depend on the RMS current, switching frequency, and dc-link voltage. The parameters involved in the loss model are categorized in Table 8.

**TABLE 8. Fixed and Identified Parameters in the Loss Model.**

| Category                         | Parameters                           |
|----------------------------------|--------------------------------------|
| Fixed (datasheet / experimental) | $R_0$ , $\alpha_R$ , $C_{oss,eq}$    |
| Identified (fit)                 | $\kappa_1$ , $\alpha_1$ , $\alpha_2$ |

## D. CALIBRATION PROCEDURE

Model calibration is carried out using calorimetric measurements over a grid of operating points ( $I_{ph,RMS}, f_{sw}$ ) at constant  $V_{dc}$  and flow-rate. The equivalent output capacitance  $C_{oss,eq}$  is extracted from zero-current switching tests by relating the measured  $E_{sw}(i_{dev} \approx 0)$  to (36).

Then, the parameters  $\kappa_1$ ,  $\alpha_1$ , and  $\alpha_2$  are jointly identified from measured data. For each operating point  $k$ , the junction temperature  $T_j^{(k)}$  is estimated from (32) using the total calorimetric loss  $P_{tot}^{(k)}$  and cold-plate temperature  $T_{cp}^{(k)}$ . The

temperature-dependent resistance  $R_{DS(on)}(T_j^{(k)})$  and conduction loss  $P_{cond,leg}^{(k)}$  are evaluated via (31) and (30), respectively, while the switching loss  $P_{sw,leg}^{(k)}$  is obtained from (37).

The parameters are determined by minimizing the least-squares error:

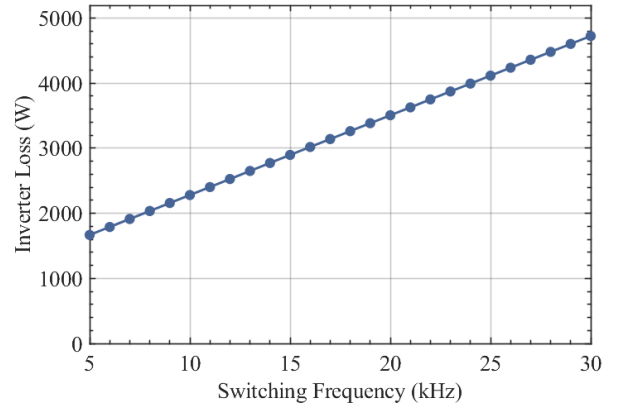
$$\min_{\kappa_1, \alpha_1, \alpha_2} \sum_k \left[ P_{tot}^{(k)} - (P_{cond,leg}^{(k)} + P_{sw,leg}^{(k)}) \right]^2, \quad (39)$$

subject to  $\kappa_1 \in (0, 1]$ ,  $\alpha_1, \alpha_2 \geq 0$ . The identified parameters enable analytical separation of conduction and switching components to validate our loss calculation framework.

## VII. CASE STUDY OF A 1.2 KV INVERTER : HOLISTIC EFFICIENCY OPTIMIZATION ACCOUNTING FOR MOTOR HARMONIC LOSSES

### A. OPTIMIZATION FRAMEWORK

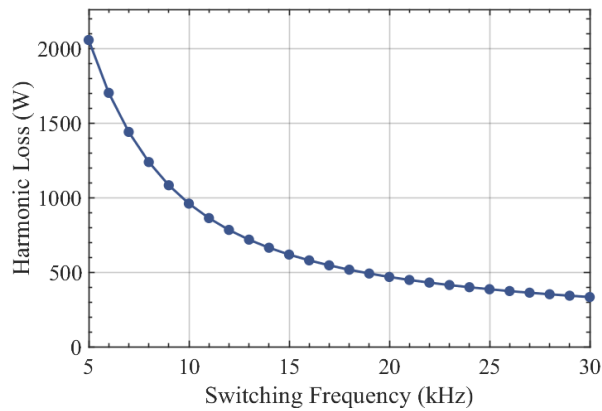
A holistic optimization procedure was conducted to identify the switching frequency range that minimizes total system losses, considering both inverter and motor contributions. The analysis was based on the Infineon FF3MR20KM1H 2 kV SiC MOSFET power module, configured with ten dies in parallel, yielding an effective die area of  $\sim 300 \text{ mm}^2$  per switch. The on-state resistance and effective output capacitance values were derived from datasheet parameters, while the switching energy was treated as a controllable function of switching speed ( $dv/dt$ ). The inverter losses over varying switching frequency are illustrated in Fig. 9.



**FIGURE 9. Inverter losses versus switching frequency at rated current.**

### B. MOTOR HARMONIC LOSSES

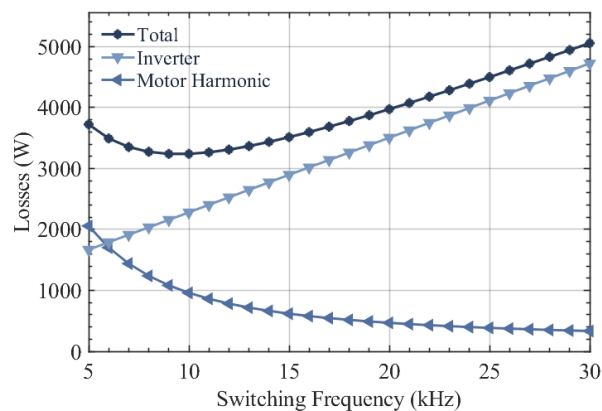
The motor harmonic losses are characterized experimentally using an impedance-based loss factor  $\Phi(\omega)$ , measured over the relevant frequency range of the traction machine for the 600 V configuration and appropriately scaled for 1200 V operation. By combining the measured  $\Phi(\omega)$  with the harmonic content of the inverter output voltage, the total harmonic losses are evaluated as a function of switching frequency. A decreasing trend in harmonic losses is observed at higher switching frequencies due to increased harmonic attenuation as illustrated in Fig. 10.



**FIGURE 10.** Motor harmonic losses calculated from impedance-based loss factor.

### C. LOSS TRADE-OFF AND OPTIMAL SWITCHING FREQUENCY

The inverter and motor loss components were combined to determine the total system losses over varying switching frequencies. This trade-off identified an optimal operating region where inverter switching losses and motor harmonic losses balance to yield the highest overall efficiency. The analysis demonstrated that increasing  $f_{sw}$  initially reduces total losses due to decreasing harmonic currents, while beyond a threshold, the inverter switching losses dominate as shown in Fig. 11.



**FIGURE 11.** Total loss trade-off between inverter and motor as a function of switching frequency.

### D. RESULTS

For the studied configuration ( $A_{die} = 300 \text{ mm}^2$ , 2 kV SiC devices, 1200 V dc-link), the optimal switching frequency is found to be approximately 10 kHz. At this operating point, the total loss reaches a minimum while remaining within the imposed thermal constraints and maintaining an acceptable contribution of motor harmonic losses.

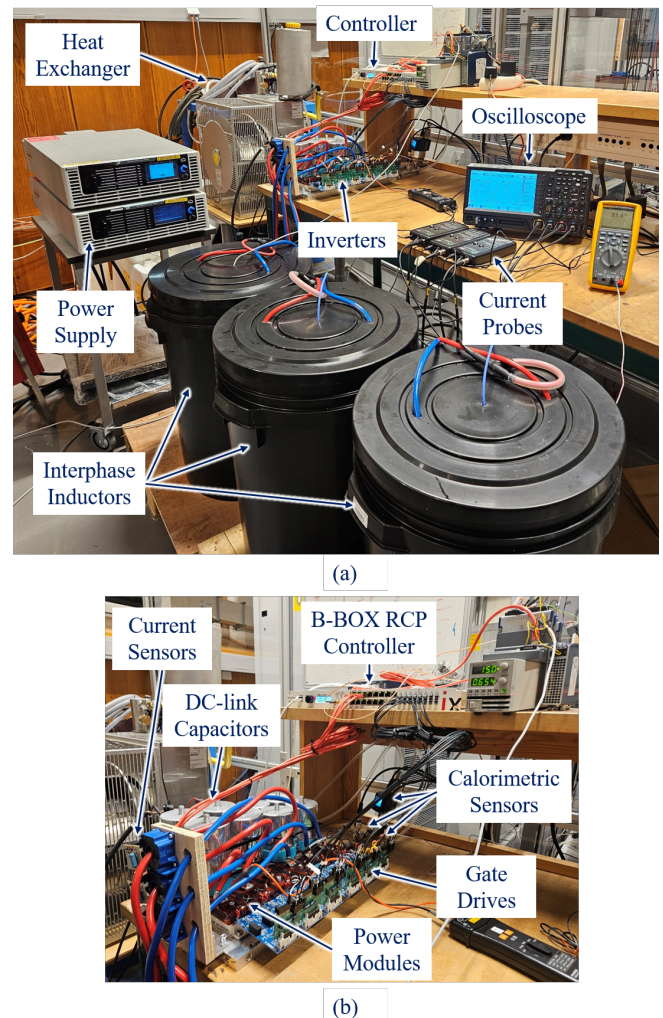
It should be noted, however, that the loss evaluation assumes a constant junction temperature across the switching-

frequency sweep. As a result, losses at lower switching frequencies may be slightly overestimated, since reduced thermal stress would lead to a lower on-state resistance  $r_{ds(on)}$  and consequently lower conduction losses in practice.

## VIII. EXPERIMENTAL VALIDATION

### A. EXPERIMENTAL SETUP

A back-to-back inverter prototype was constructed to experimentally validate the proposed analytical loss calculation, which is given in Fig. 12.



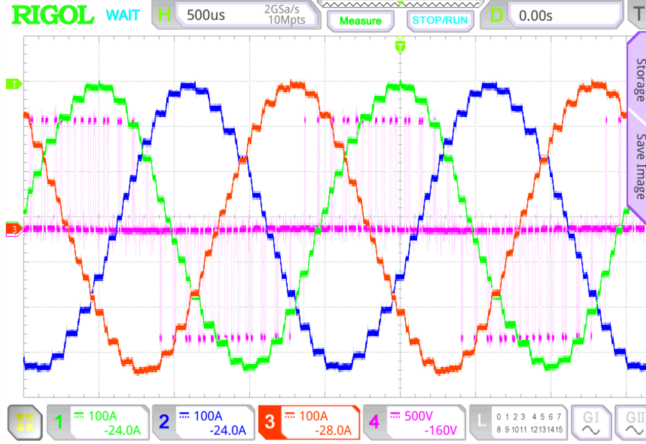
**FIGURE 12.** Experimental setup employed for validation. a) Back-to-back inverters prototype. b) Inverters under test.

The setup consists of two identical two-level three-phase voltage-source inverters built with Infineon FF3MR20KM1H 2 kV SiC MOSFET power modules, operating from a 1200 V dc-link. Both inverters are mounted on liquid-cooled cold plates with regulated coolant flow, and the system is controlled through an Imperix B-Box RCP platform implementing PWM control and real-time data acquisition. The setup includes high-voltage differential probes and isolated current sensors for monitoring dc-link voltage and phase currents.

The cold-plate inlet and outlet temperatures, coolant flow rate, and total calorimetric heat dissipation are continuously measured to determine total inverter losses.

### B. MEASUREMENT PROCEDURE

First, time domain waveforms of a phase voltage and phase currents were recorded to verify the rated operation of the prototype inverter, shown in Fig. 13.



**FIGURE 13.** Measured a phase voltage and three-phase current waveforms under steady-state operation at  $m_a = 1$ ,  $f_{sw} = 10\text{kHz}$ ,  $I_{ph} = 200\text{A}_{RMS}$

Second, calorimetric efficiency measurements were conducted under steady-state thermal conditions by operating the inverter at various RMS phase currents and switching frequencies. At each operating point, the dc-link voltage, phase current, and switching frequency were kept constant while monitoring electrical input and output powers for verification. The RMS phase current was incrementally varied from 50 A to 200 A, and the switching frequency was sampled between 7 kHz and 15 kHz. Each operating point was maintained until thermal equilibrium was reached to ensure repeatable calorimetric data.

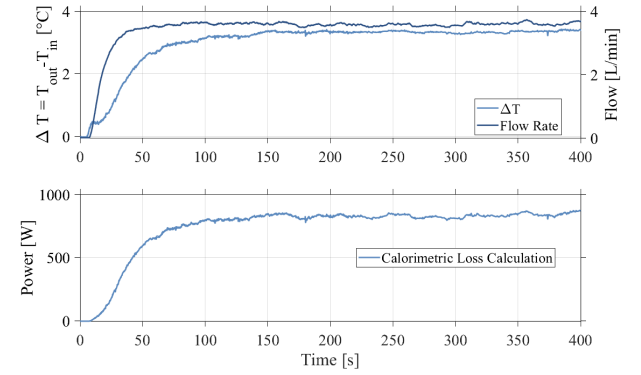
### C. DATA ACQUISITION

An example measurement for flow rate, and temperature differences between inlet and outlet are plotted in Fig. 14.

Zero-current switching losses were measured at two different switching frequencies, as presented in Table 9. In addition, steady-state total losses and cold-plate temperatures were obtained over a range of phase currents and switching frequencies, presented in Table 10 and Table 11, respectively.

**TABLE 9.** Measured total loss  $P_{loss}$  (W) at  $I_{RMS} = 0$  at two switching frequencies for 6-legs.

| $I_{RMS}$ (A) ↓ / $f_{sw}$ (kHz) → | 7      | 19     |
|------------------------------------|--------|--------|
| 0                                  | 298.61 | 680.82 |



**FIGURE 14.** Example data of the measured flow rate ( $m$ ) and temperature differences ( $T_{out} - T_{in}$ ), and calculated loss for 6-legs.

**TABLE 10.** Measured total loss  $P_{loss}$  (W) vs.  $I_{RMS}$  and  $f_{sw}$  for 6-legs.

| $I_{RMS}$ (A) ↓ / $f_{sw}$ (kHz) → | 7      | 10     | 15     | 19     |
|------------------------------------|--------|--------|--------|--------|
| 50                                 | 636.9  | 998.1  | 1204.5 | 1549.8 |
| 100                                | 1382.9 | 2037.2 | 2398.1 | 3361.3 |
| 150                                | 2307.9 | 2942.9 | 3861.7 | 5067.7 |
| 200                                | 3479.5 | 4417.1 | 5948.9 | 7878.4 |

**TABLE 11.** Measured cold-plate temperature  $T_{cp}$  (°C) vs.  $I_{RMS}$  and  $f_{sw}$

| $I_{RMS}$ (A) ↓ / $f_{sw}$ (kHz) → | 7      | 10     | 15     | 19     |
|------------------------------------|--------|--------|--------|--------|
| 50                                 | 27.618 | 26.929 | 28.869 | 29.638 |
| 100                                | 29.669 | 30.652 | 33.661 | 30.357 |
| 150                                | 33.738 | 35.472 | 38.705 | 42.314 |
| 200                                | 37.071 | 39.829 | 46.275 | 53.120 |

### D. RESULTS

The equivalent output capacitance  $C_{oss,eq}$  is identified using zero-current loss measurements, and the model parameters  $\{\kappa_1, \alpha_1, \alpha_2\}$  are estimated by using total losses and cold plate temperatures over varying current and switching frequency. The resulting parameters, summarized in Table 12, capture the combined electro-thermal behavior of the SiC devices and the cooling interface under the applied conditions.

**TABLE 12.** Identified Parameters from calorimetric Measurements.

| Parameter                     | Symbol       | Identified Value          |
|-------------------------------|--------------|---------------------------|
| Thermal coupling coefficient  | $\kappa_1$   | 0.0192 K/W                |
| Linear current coefficient    | $\alpha_1$   | 172.1 $\mu\text{J/A}$     |
| Quadratic current coefficient | $\alpha_2$   | 151.2 $\text{nJ/A}^2$     |
| Equivalent output capacitance | $C_{oss,eq}$ | 3.66 nF                   |
| On-state resistance at 25°    | $R_0$        | 3.2 $\Omega$              |
| Thermal coefficient           | $\alpha_r$   | 0.0115 $1/^\circ\text{C}$ |

Table 13 presents the modeled zero-current switching losses using the identified equivalent output capacitance and the corresponding relative errors. Table 14 summarizes the



modeled total inverter losses and their relative errors with respect to experimental measurements.

**TABLE 13.** Modeled loss  $P_{\text{model}}$  (W) (top) and relative error (%) (bottom) at  $I_{\text{RMS}} = 0$  for 6-legs.

| $I_{\text{RMS}}$ (A) ↓ / $f_{\text{sw}}$ (kHz) → | 7                     | 19                    |
|--------------------------------------------------|-----------------------|-----------------------|
| 0                                                | 310.0 (W)<br>3.82 (%) | 702.0 (W)<br>3.12 (%) |

**TABLE 14.** Modelled loss  $P_{\text{model}}$  (W) (top) and relative error (%) (bottom) for 6-legs.

| $I_{\text{RMS}}$ (A) ↓ / $f_{\text{sw}}$ (kHz) → | 7               | 10               | 15              | 19              |
|--------------------------------------------------|-----------------|------------------|-----------------|-----------------|
| 50                                               | 622.5<br>−2.25  | 866.6<br>−13.18  | 1275.5<br>5.90  | 1692.3<br>3.42  |
| 100                                              | 1284.7<br>−7.10 | 1739.3<br>−14.63 | 2505.4<br>4.48  | 3250.1<br>−3.31 |
| 150                                              | 2188.8<br>−5.16 | 2900.8<br>−1.43  | 3809.4<br>−1.35 | 5047.8<br>−0.39 |
| 200                                              | 3428.6<br>−1.29 | 4467.7<br>1.15   | 6213.1<br>4.44  | 7630.7<br>−3.14 |

Consequently, Table 15 separates the modeled inverter losses into conduction and switching components.

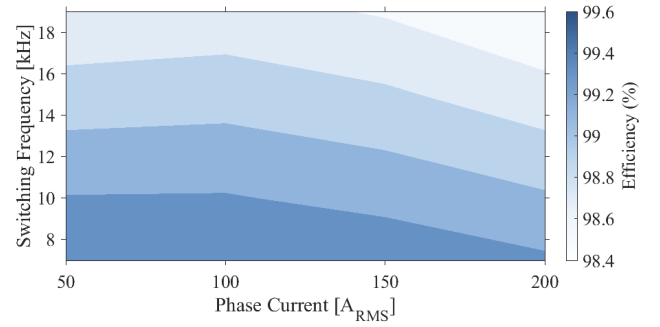
**TABLE 15.** Modelled conduction loss  $P_{\text{cond}}$  (W) (top) and switching loss  $P_{\text{sw}}$  (W) (bottom) for 6-legs.

| $I_{\text{RMS}}$ (A) ↓ / $f_{\text{sw}}$ (kHz) → | 7                | 10               | 15               | 19               |
|--------------------------------------------------|------------------|------------------|------------------|------------------|
| 50                                               | 59.0<br>563.5    | 61.6<br>805.0    | 68.0<br>1207.5   | 68.0<br>1624.3   |
| 100                                              | 275.0<br>1009.7  | 296.8<br>1442.5  | 341.7<br>2163.7  | 341.7<br>2908.4  |
| 150                                              | 721.4<br>1467.4  | 804.5<br>2096.3  | 804.5<br>3004.9  | 1064.9<br>3982.9 |
| 200                                              | 1498.2<br>1936.4 | 1491.1<br>2976.6 | 2063.3<br>4149.8 | 2374.6<br>5256.1 |

## E. INTERPRETATION AND DISCUSSION

The measured total losses closely followed the combined conduction- and switching-loss trends predicted by the analytical model, confirming the validity of the parameter identification procedure. A close agreement is observed over the entire operating range, indicating that the proposed electro-thermal model successfully captures both the conduction- and switching-loss mechanisms under real inverter operating conditions.

Fig. 15 presents the measured inverter efficiency as a function of RMS phase current. At full-load operation ( $P_{\text{out}} = 250$  kW,  $V_{\text{dc}} = 1200$  V,  $f_{\text{sw}} = 10$  kHz), the measured efficiency reaches 99.12%, which lies within an approximately 10% error margin of the analytical loss prediction. This agreement verifies that the proposed modeling and parameter extraction approach can accurately separate and quantify the



**FIGURE 15.** Measured inverter efficiency along with RMS phase current and switching frequency at  $V_{\text{dc}} = 1200$  V.

conduction and switching loss components, thereby providing a reliable foundation for an efficiency optimization framework. Further, these results confirm that calorimetric validation enables reliable loss measurement without intrusive high-bandwidth voltage and current probes.

Further, the volumes of the manufactured inverter components were measured, as reported in Table 16, to validate the virtual prototyping method. The measured values show good agreement with the estimated volumes, with average error within 11%.

**TABLE 16.** Component Volumes, Specific Power Density, and Cubic Power Density for Established Inverter.

| Power Module (L) | Dc-link Capacitor (L) | Cold Plate (L) | Aux. System (L) | $\rho_p$ (kW/L) | $\rho_c$ (kW/L) |
|------------------|-----------------------|----------------|-----------------|-----------------|-----------------|
| 0.56             | 3.56                  | 0.62           | 0.32            | 49.4            | 20.83           |

## IX. LIMITATIONS AND FUTURE WORK

This study focused on the effect of raising voltage using a two-level three-phase SiC traction inverter at a rated 250 kW operating point with equal-loss constraints, using a single machine geometry and sinusoidal PWM. Drive-cycle effects, partial-load efficiency [36], lifetime/aging of dc-link capacitors, gate-driver dynamics across wider  $dv/dt$ , and alternative topologies (e.g., multi-phase, multi-level inverters[12]) were out of the scope, and future work may

- (i) extend the framework to multi-level multi-phase inverters and wide-range  $f_{\text{sw}}/A_{\text{die}}$  co-optimization,
- (ii) include drive-cycle weighted efficiency,
- (iii) integrate capacitor lifetime models and reliability constraints.

## X. CONCLUSION

A system-level assessment of increasing the dc-link voltage from 800 V to 1200 V for heavy-duty electric vehicle powertrains was conducted, encompassing battery and charging performance, inverter loss–volume trade-offs, motor harmonic behavior, and experimentally calibrated loss modeling. The analysis and experimental validation indicate that a higher dc-link voltage reduces current levels and improves system



compactness while maintaining comparable inverter losses and thermal feasibility under equal-loss operating conditions. Key outcomes are summarized as follows:

- Increasing dc-link voltage from 800 V to 1200 V can lower either cable losses or mass by approximately 56% and 33%, respectively; further enabling fast charging and a full recharge in about 36 min for 600 kWh battery packs.
- Equal-loss inverter designs exhibit similar thermal behavior; however, the 1200 V case achieves higher power density (experimentally 49.4 kW/L specific power density, and 20.83 kW/L cubic power density) due to reduced dc-link capacitance volume.
- Measured inverter efficiency of 99.12% agrees within 10% of analytical loss predictions, confirming model accuracy.

Overall, increasing the dc-link voltage to 1200 V offers a practical and standards-compliant route toward more compact, efficient, and fast-charging heavy-duty electric powertrains, while preserving comparable inverter losses and thermal performance under consistent design constraints.

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